

```
In [1]: import matplotlib.pyplot as plt
from factor_diffusion_sample import generate_rejection
import matplotlib.patches as mpatches
from matplotlib.colors import TwoSlopeNorm
from factor_model import *
from sklearn.decomposition import PCA
from scipy.stats import gaussian_kde
from scenario_generator import *
import numpy as np

with open("cfg.yaml") as f:
    _exp = yaml.safe_load(f)["experiment_name"]
PREFIX = f"model/{_exp}"
with open(f"{PREFIX}/cfg.yaml") as f:
    _cfg = yaml.safe_load(f)

FACTOR_NAMES = _cfg["factor_names"]
NUM_GENERATE = _cfg["num_generate"]
print(f"experiment: {_exp}")
```

```
/home/renyi/anaconda3/envs/diffusion_factor_model/lib/python3.12/site-packages/tqdm/auto.py:21: TqdmWarning: IProgress not found. Please update jupyter and ipywidgets. See https://ipywidgets.readthedocs.io/en/stable/user_install.html
```

```
from .autonotebook import tqdm as notebook_tqdm
experiment: DDPM
```

```
In [2]: def plot_vol_panel(ax, xlim, title_suffix, datasets, fidx, threshold=None):
    for label, data, color in datasets:
        col = data[:, fidx] if data.ndim == 2 else data
        clipped = col[(col >= xlim[0]) & (col <= xlim[1])]
        if len(clipped) < 2:
            continue
        ax.hist(clipped, bins=40, density=True, alpha=0.55,
                color=color, label=label, edgecolor="white", linewidth=0.4)
        kde = gaussian_kde(col, bw_method=0.3)
        x_grid = np.linspace(xlim[0], xlim[1], 400)
        ax.plot(x_grid, kde(x_grid), color=color, linewidth=2)

    if threshold is not None:
        ax.axvline(threshold, color="black", linestyle="--", linewidth=1.5,
                  label=f"threshold = {threshold:.4f}")
    ax.set_xlabel(FACTOR_NAMES[fidx])
    ax.set_ylabel("density")
    ax.set_xlim(xlim)
    ax.set_title(title_suffix)
    ax.legend(fontsize=8, framealpha=0.9)
    for i, (label, data, color) in enumerate(datasets):
        col = data[:, fidx] if data.ndim == 2 else data
        pct = (col <= threshold).mean() * 100 if threshold is not None else 0
        visible = ((col >= xlim[0]) & (col <= xlim[1])).sum()
        ax.text(0.02, 0.97 - i * 0.10,
               f"{label}: {pct:.0f}% <= thr ({visible}/{len(col)} visible)
```

```
transform=ax.transAxes, fontsize=8,
color=color, va="top")
```

- max is increasing via sample size and it is not stable in market factor
- N = 4096 other fatcor are stable

```
In [3]: DEVICE          = torch.device("cuda" if torch.cuda.is_available() else "cpu")
NUM_TIMESTEPS = _cfg["num_timesteps"]
LEVY_ALPHA    = _cfg["levy_alpha"]
BATCH_SIZE    = _cfg["batch_size"]
FACTOR_DIM    = len(_cfg["factor_names"])
CHECKPOINT    = f"{PREFIX}/checkpoints/{_cfg['ckpt_name']}.pt"
print(CHECKPOINT)
```

model/DDPM/checkpoints/DLPM_L.pt

```
In [4]: # util functions
def _to_norm(scaler, value, idx):
    dummy = np.zeros((1, FACTOR_DIM))
    dummy[0, idx] = value
    return float(scaler.transform(dummy)[0, idx])

def make_cond_fn(interval, scaler, feature_idx=0, normalize=True):
    thl, thr = interval

    if normalize:
        qmax_norm = _to_norm(scaler, thr, feature_idx)

        def cond_fn(x0_hat):
            return torch.relu(x0_hat[:, feature_idx] - qmax_norm) ** 2
    else:
        def cond_fn(x0_hat):
            return torch.relu(x0_hat[:, feature_idx] - thr) ** 2

    cond_fn.label = f"{FACTOR_NAMES[feature_idx]} < {thr:.4f}"

    return cond_fn
```

```
In [5]: # load achievements
"""
ckpt = torch.load(CHECKPOINT, map_location=DEVICE, weights_only=False)
model = FactorDenoiser(**ckpt["model_kwargs"]).to(DEVICE)
model.load_state_dict(ckpt["model_state"])
gammas, bargammas, sigmas, barsigmas = levy_noise_schedule(LEVY_ALPHA, NUM_T
scaler = ckpt["scaler"]
args = (model, gammas, bargammas, sigmas, barsigmas, LEVY_ALPHA, scaler)

"""
```

```
Out[5]: '\nckpt = torch.load(CHECKPOINT, map_location=DEVICE, weights_only=False)\n
model = FactorDenoiser(**ckpt["model_kwargs"]).to(DEVICE)\nmodel.load_state
_dict(ckpt["model_state"])\ngammas, bargammas, sigmas, barsigmas = levy_noi
se_schedule(LEVY_ALPHA, NUM_TIMESTEPS)\nscaler = ckpt["scaler"]\nargs =
(model, gammas, bargammas, sigmas, barsigmas, LEVY_ALPHA, scaler)\n\n'
```

```
In [6]: """
Ns = [128, 256, 512, 1028, 2048, 4096, 10000]
uncons = [generate(*args, num_samples=N) for N in Ns]
maxns = [np.max(uncon[0], axis=0) for uncon in uncons]
maxns
q3s    = [np.percentile(uncon[0], 3, axis=0) for uncon in uncons]
qmins = [uncon[0].min(axis=0) for uncon in uncons]
fidx  = FACTOR_NAMES.index("momentum")
conds = []
for q3, qmin in zip(q3s, qmins):
    cond = make_cond_fn((qmin[fidx], q3[fidx]), scaler, fidx)
    conds.append(cond)

cons = [generate(*args, cond_fn=c, guidance_scale=GUIDANCE_S, num_samples=r
              for c, n in zip(conds, Ns)]
maxns = [np.max(con[0], axis) for con in cons]
maxns
"""
```

```
Out[6]: '\nNs = [128, 256, 512, 1028, 2048, 4096, 10000]\nuncons = [generate(*args,
num_samples=N) for N in Ns]\nmaxns = [np.max(uncon[0], axis=0) for uncon in un
cons]\nmaxns\nq3s    = [np.percentile(uncon[0], 3, axis=0) for uncon in un
cons]\nqmins = [uncon[0].min(axis=0) for uncon in uncons]\nfidx = FACTOR_N
AMES.index("momentum")\nconds = []\nfor q3, qmin in zip(q3s, qmins):\n    c
ond = make_cond_fn((qmin[fidx], q3[fidx]), scaler, fidx)\n    conds.append
(cond)\n\ncons = [generate(*args, cond_fn=c, guidance_scale=GUIDANCE_S, nu
m_samples=n)\n                for c, n in zip(conds, Ns)]\nmaxns = [np.max(con[0],
axis) for con in cons]\nmaxns\n'
```

```
In [7]: # unconditional sample and conditional config
GUIDANCE_S = 5 # guidance scale
N = 4096 # sample number
train_model = load_model(f"{PREFIX}")
diffusionScenario = ScenarioGenerator(train_model, DiffusionSampler(CHECKPOINT))
```

```
In [8]: uncon = diffusionScenario.factor_generate(N)
uncon.shape
```

```
Out[8]: (4096, 7)
```

```
In [9]: #momentum < 3% quantile
q3      = np.percentile(uncon, 3, axis=0)
qmin    = uncon.min(axis=0)
fidx    = FACTOR_NAMES.index("volatility")

lower003_fn = make_cond_fn((qmin[fidx], q3[fidx]), diffusionScenario.sample
lower003 = diffusionScenario.cond_factor_generate(N, lower003_fn)
print(lower003_fn.label)
```

```
volatility < -0.0074
```

```
In [10]: lower003_fn.label
```

```
Out[10]: 'volatility < -0.0074'
```

```
In [11]: gaussian_model = load_model("model/Gaussian")
gaussianScenario = ScenarioGenerator(gaussian_model, GaussianSampler(gaussian_model))
sampled_gaussian = gaussianScenario.cond_factor_generate(N, lower003_fn)
```

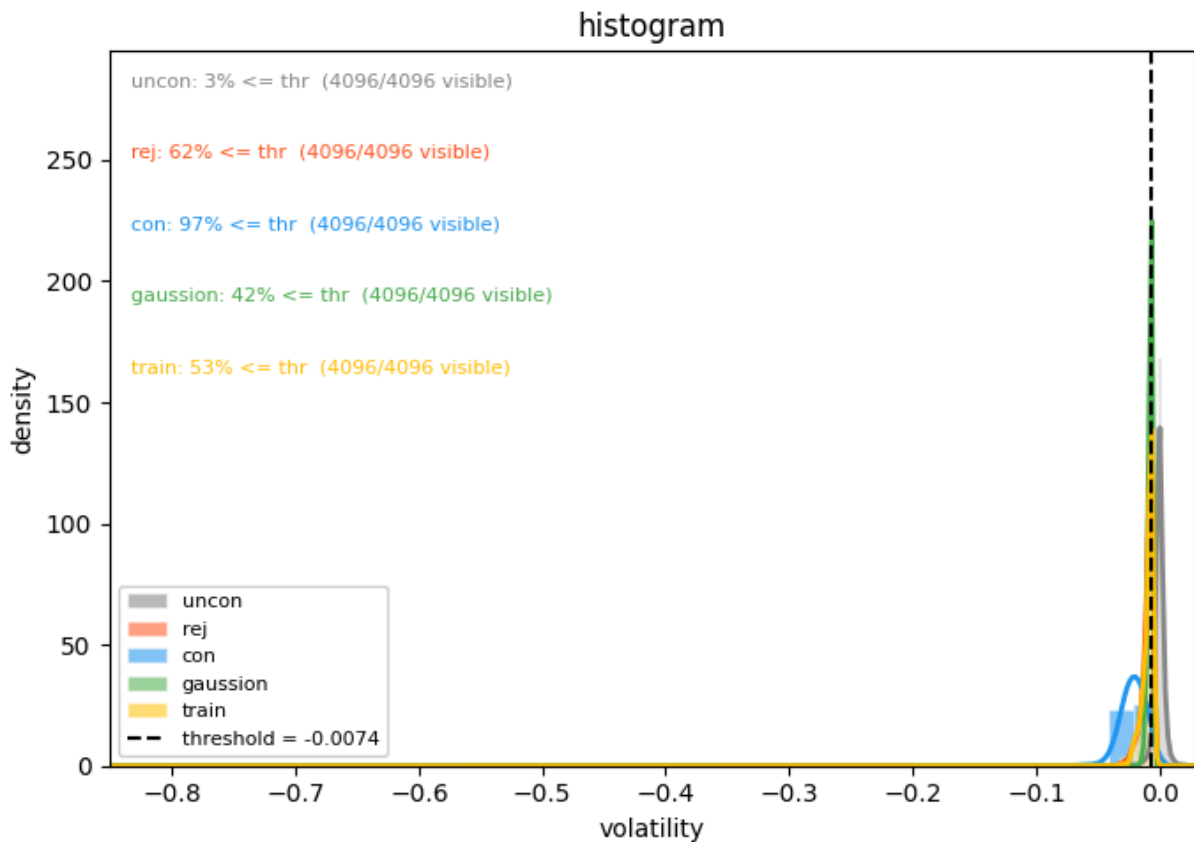
```
In [12]: resampledScenario = ScenarioGenerator(train_model, ResampleSampler(train_model))
sampled_train = resampledScenario.cond_factor_generate(N, lower003_fn)
```

```
In [13]: REJ_CACHE = f"{PREFIX}/samples/rej__vol_{_cfg['ckpt_name']}_N{N}_s{GUIDANCE_SCALE}"
if os.path.exists(REJ_CACHE):
    rej = np.load(REJ_CACHE)
    print(f"Loaded cached rej samples from {REJ_CACHE} shape={rej.shape}")
else:
    print("Cache not found, running rejection sampling (slow)...")
    rej = generate_rejection(diffusionScenario.sampler.model, diffusionScenario,
                             num_samples=N, guidance_scale=GUIDANCE_SCALE, L = cfg['L'])
    np.save(REJ_CACHE, rej)
    print(f"Saved to {REJ_CACHE} shape={rej.shape}")
```

Loaded cached rej samples from model/DDPM/samples/rej__vol_DLPM_L_N4096_s5.npy shape=(4096, 7)

```
In [14]: COLORS = {
    "uncon": "#888888",
    "con": "#2196F3",
    "rej": "#FF5722",
    "gaussian": "#4CAF50",
    "train": "#FFC107"
}
```

```
In [15]: datasets_hist = [
    ("uncon", uncon, COLORS["uncon"]),
    ("rej", rej, COLORS["rej"]),
    ("con", lower003, COLORS["con"]),
    ("gaussian", sampled_gaussian, COLORS["gaussian"]),
    ("train", sampled_train, COLORS["train"]),
]
all_data = np.concatenate([d[:, fidx] for _, d, _ in datasets_hist])
full_lim = (all_data.min() - abs(all_data.min()) * 0.05,
            all_data.max() + abs(all_data.max()) * 0.05)
fig2, ax_full = plt.subplots(figsize=(7, 5))
plot_vol_panel(ax_full, full_lim, "histogram", datasets_hist, fidx=fidx, threshold=0.05)
fig2.tight_layout()
plt.show()
```



```
In [16]: datasets = [
    ("uncon", uncon, COLORS["uncon"]),
    # ("rej", rej, "#FF5722"),
    ("con", lower003, COLORS["con"]),
]
x_center = np.arange(FACTOR_DIM)
offsets = [-0.25, 0.0, 0.25]
fig, ax = plt.subplots(figsize=(13, 5))

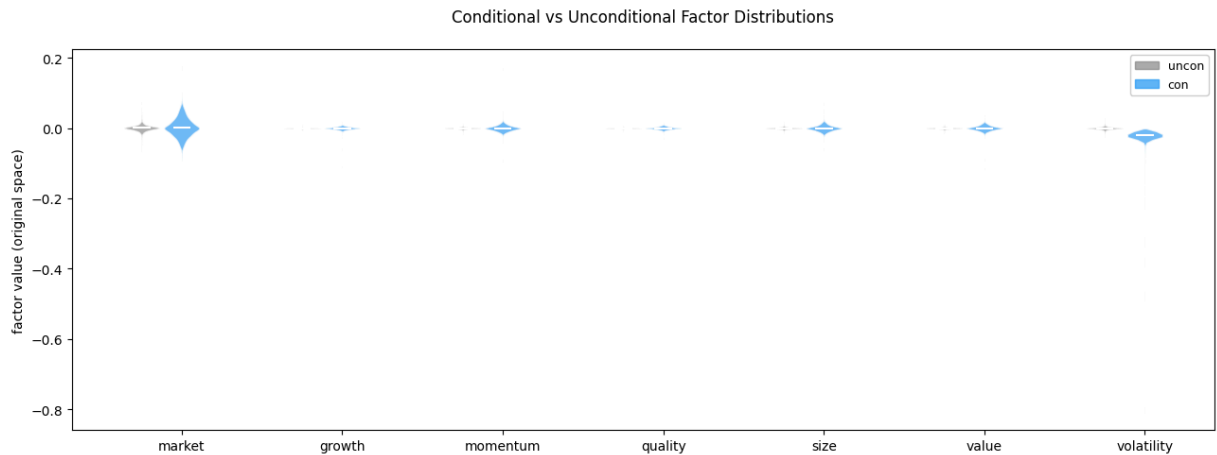
for offset, (label, data, color) in zip(offsets, datasets):
    parts = ax.violinplot(
        [data[:, i] for i in range(FACTOR_DIM)],
        positions=x_center + offset,
        widths=0.22,
        showmedians=True,
        showextrema=False,
    )
    for pc in parts["bodies"]:
        pc.set_facecolor(color)
        pc.set_edgecolor("white")
        pc.set_alpha(0.65)
    parts["cmedians"].set_color("white")
    parts["cmedians"].set_linewidth(1.5)

ax.set_xticks(x_center)
ax.set_xticklabels(FACTOR_NAMES, fontsize=10)
ax.set_ylabel("factor value (original space)")
ax.set_title(
    f"Conditional vs Unconditional Factor Distributions\n"
```

```
)

handles = [mpatches.Patch(color=c, alpha=0.7, label=l) for l, _, c in dataset]
ax.legend(handles=handles, fontsize=9, framealpha=0.9)

fig.tight_layout()
plt.show()
```



Correlation Heatmaps - Unconditional vs Conditional

```
In [17]: def corr_mat(data):
          return np.corrcoef(data, rowvar=False)

corr_uncon = corr_mat(uncon)
corr_rej = corr_mat(rej)
corr_low = corr_mat(lower003)
corr_gaussian = corr_mat(sampled_gaussian)
corr_train = corr_mat(train_model.F.drop(columns=['alpha']).values)
```

```
In [18]: # --- load historical factor data ---
hist = pd.read_csv(f"{PREFIX}/factors.csv", index_col=0, parse_dates=True)
# drop alpha column if present
hist = hist[[c for c in hist.columns if c in FACTOR_NAMES]]
hist = hist[FACTOR_NAMES] # align column order

# --- filter historical days matching condition: momentum < q3[fidx] ---
mom_thresh = q3[fidx] # same threshold used to generate lower003
hist_stress = hist[hist.iloc[:, fidx] < mom_thresh].values
gen_stress = lower003 # (N, FACTOR_DIM)

print(f"Historical stress days: {len(hist_stress)}")
print(f"Generated stress samples: {len(gen_stress)}")
print(f"Momentum threshold: {mom_thresh:.4f}")
```

```
Historical stress days: 177
Generated stress samples: 4096
Momentum threshold: -0.0074
```

```
In [19]: methods_to_plot = [
          ("Historical", hist_stress, "#FF5722", "-", 0.35),
          ("Diffusion", lower003, COLORS["con"], "-", 0.25),
```

```

        ("Rejection",      rej,      COLORS["rej"],  "--", 0.20),
        ("Gaussian",      sampled_gaussian, COLORS["gaussian"], "--", 0.20),
    ]

    uncon_label, uncon_data, uncon_color, uncon_ls, uncon_alpha = ("Uncon", uncon_data, "black", "solid", 0.2)

    nrows = len(methods_to_plot)
    ncols = FACTOR_DIM

    fig, axes = plt.subplots(nrows, ncols, figsize=(ncols * 2.5, nrows * 2.2),
                             sharex="col", sharey="col")

    all_datasets = [d for _, d, _, _ in methods_to_plot] + [uncon_data]
    x_grids = []
    for i in range(ncols):
        cols = [d[:, i] for d in all_datasets]
        xlo = min(np.percentile(c, 0.5) for c in cols)
        xhi = max(np.percentile(c, 99.5) for c in cols)
        pad = (xhi - xlo) * 0.05
        x_grids.append(np.linspace(xlo - pad, xhi + pad, 400))

    for r, (label, data, color, ls, alpha) in enumerate(methods_to_plot):
        for i, name in enumerate(FACTOR_NAMES):
            ax = axes[r, i]
            x_grid = x_grids[i]

            col_uncon = uncon_data[:, i]
            kde_uncon = gaussian_kde(col_uncon, bw_method=0.3)
            ax.fill_between(x_grid, kde_uncon(x_grid), alpha=uncon_alpha, color=uncon_color)
            ax.plot(x_grid, kde_uncon(x_grid), color=uncon_color, linewidth=1.5, linestyle=uncon_ls)

            col_method = data[:, i]
            kde_method = gaussian_kde(col_method, bw_method=0.3)
            ax.fill_between(x_grid, kde_method(x_grid), alpha=alpha, color=color)
            ax.plot(x_grid, kde_method(x_grid), color=color, linewidth=1.5, linestyle=ls)

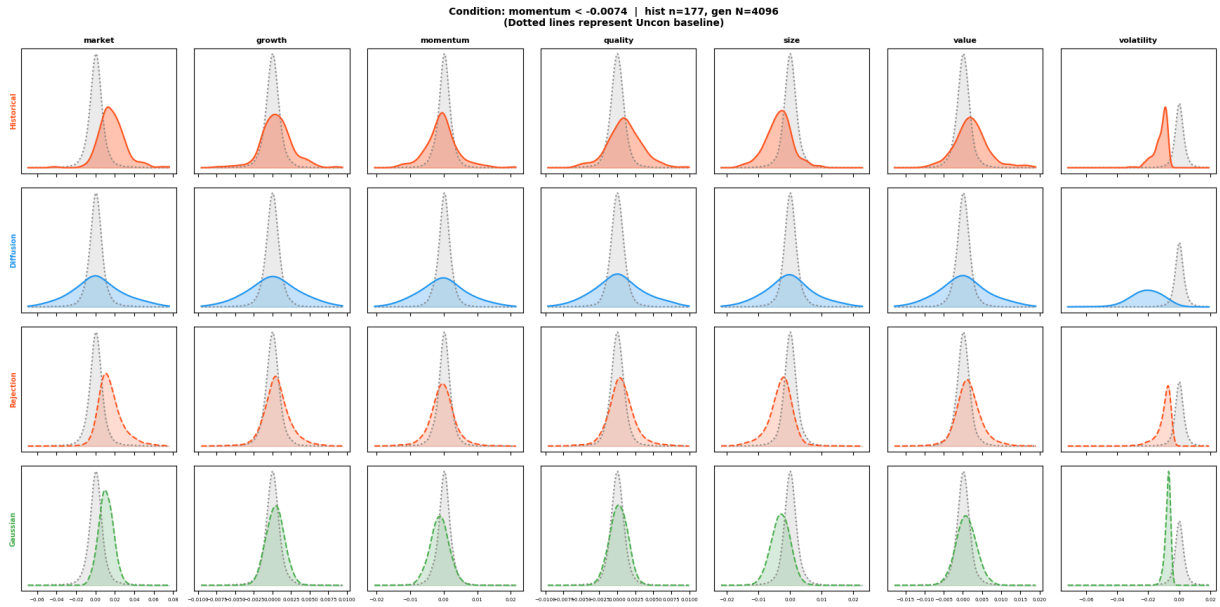
            ax.set_yticks([])
            ax.tick_params(labelsize=5)

            if r == 0:
                ax.set_title(name, fontsize=8, fontweight="bold")
            if i == 0:
                ax.set_ylabel(label, fontsize=7, color=color, fontweight="bold")

    fig.suptitle(
        f"Condition: momentum < {mom_thresh:.4f} | hist n={len(hist_stress)},",
        f"(Dotted lines represent Uncon baseline)",
        fontsize=10, fontweight="bold"
    )

```

```
plt.tight_layout()
plt.show()
```



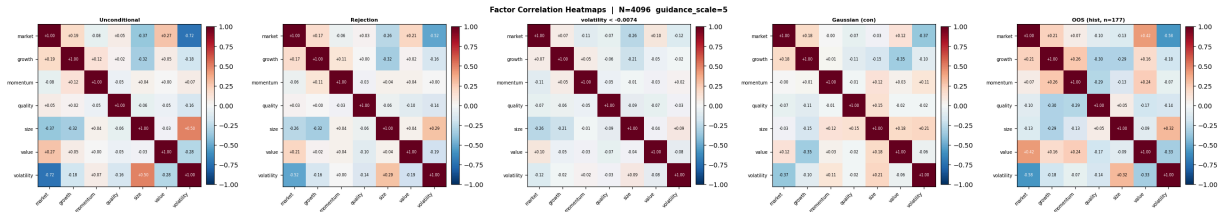
```
In [20]: corr_hist = corr_mat(hist_stress)

all_panels = [
    ("Unconditional",    corr_uncon),
    ("Rejection",       corr_rej),
    (lower003_fn.label, corr_low),
    ("Gaussian (con)",   corr_gaussian),
    (f"00S (hist, n={len(hist_stress)})", corr_hist),
]

norm = TwoSlopeNorm(vmin=-1, vcenter=0, vmax=1)
fig, axes = plt.subplots(1, 5, figsize=(26, 4.5), constrained_layout=True)

for ax, (name, corr) in zip(axes, all_panels):
    im = ax.imshow(corr, cmap="RdBu_r", norm=norm, aspect="equal")
    ax.set_xticks(range(FACTOR_DIM))
    ax.set_yticks(range(FACTOR_DIM))
    ax.set_xticklabels(FACTOR_NAMES, rotation=45, ha="right", fontsize=7)
    ax.set_yticklabels(FACTOR_NAMES, fontsize=7)
    ax.set_title(name, fontweight="bold", fontsize=8, pad=4)
    for ii in range(FACTOR_DIM):
        for jj in range(FACTOR_DIM):
            ax.text(jj, ii, f"{corr[ii,jj]:+.2f}",
                    ha="center", va="center", fontsize=5.5,
                    color="white" if abs(corr[ii,jj]) > 0.4 else "black")
    plt.colorbar(im, ax=ax, fraction=0.046, pad=0.03)

fig.suptitle(
    f"Factor Correlation Heatmaps | N={N} guidance_scale={GUIDANCE_S}",
    fontsize=11, fontweight="bold"
)
plt.show()
```

In [21]: `import itertools`

```
idx_pairs = list(itertools.combinations(range(FACTOR_DIM), 2))
pair_labels = [f"{FACTOR_NAMES[i][:3]}-{FACTOR_NAMES[j][:3]}" for i, j in idx_pairs]

def upper_tri(corr):
    return np.array([corr[i, j] for i, j in idx_pairs])

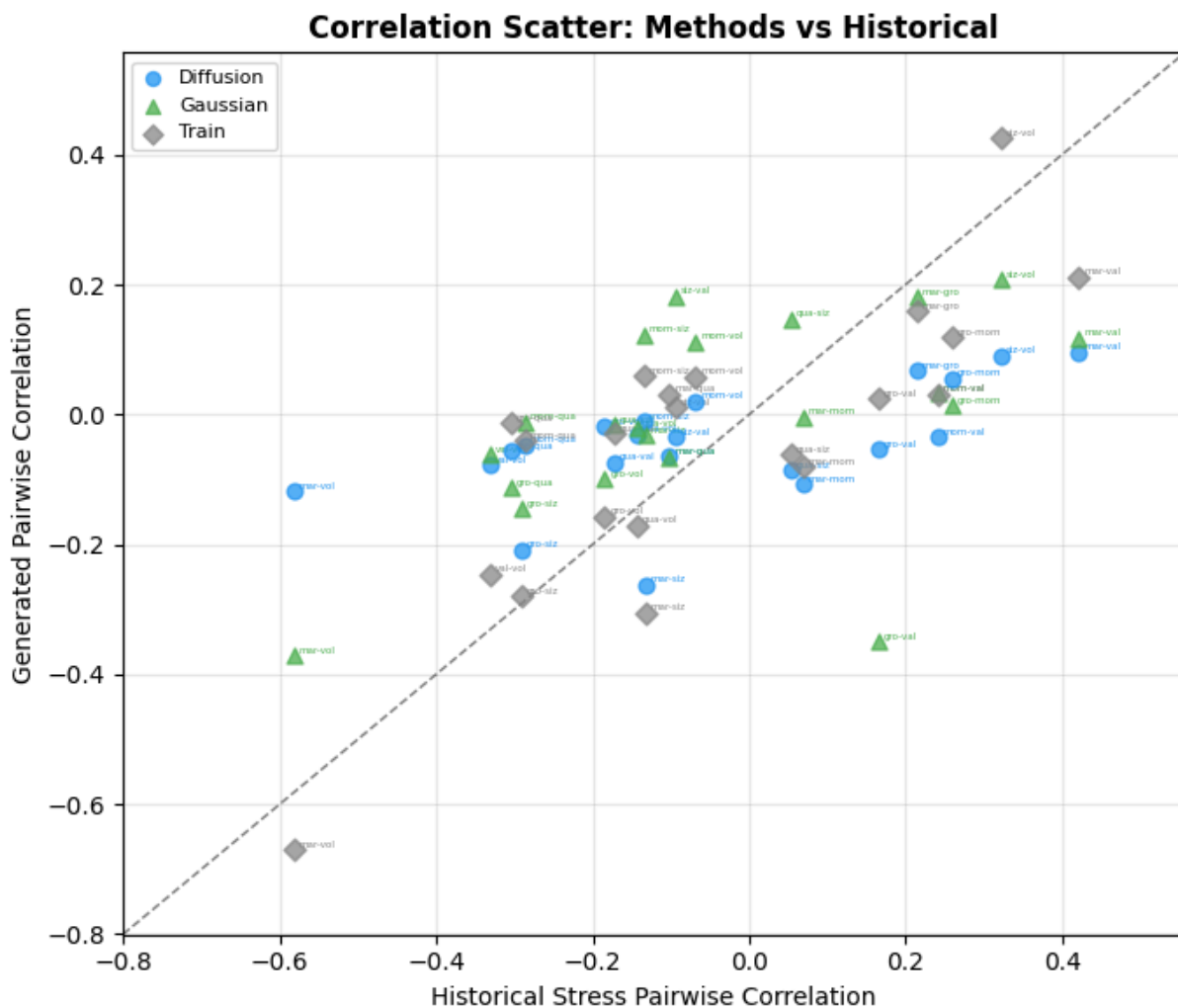
hist_vec = upper_tri(corr_hist)
methods = [
    ("Diffusion", upper_tri(corr_low), COLORS["con"], "o"),
    # ("Rejection", upper_tri(corr_rej), COLORS["rej"], "s"),
    ("Gaussian", upper_tri(corr_gaussian), COLORS["gaussian"], "^"),
    ("Train", upper_tri(corr_train), COLORS["uncon"], "D")
]

all_vecs = [hist_vec] + [v for _, v, _, _ in methods]
lim = [min(v.min() for v in all_vecs), max(v.max() for v in all_vecs)]
pad = (lim[1] - lim[0]) * 0.12 or 0.05
lim = [lim[0] - pad, lim[1] + pad]

fig, ax = plt.subplots(figsize=(7, 6))
ax.plot(lim, lim, "k--", lw=1, alpha=0.5)

for label, vec, color, mk in methods:
    ax.scatter(hist_vec, vec, color=color, s=40, marker=mk,
               zorder=4, label=label, alpha=0.75)
    for k, lbl in enumerate(pair_labels):
        ax.annotate(lbl, (hist_vec[k], vec[k]),
                    fontsize=4, xytext=(2, 1),
                    textcoords="offset points", color=color)

ax.set_xlim(lim); ax.set_ylim(lim)
ax.set_xlabel("Historical Stress Pairwise Correlation", fontsize=10)
ax.set_ylabel("Generated Pairwise Correlation", fontsize=10)
ax.set_title("Correlation Scatter: Methods vs Historical", fontweight="bold")
ax.legend(fontsize=8, markerscale=0.9, handlelength=1.2)
ax.grid(True, alpha=0.3)
plt.tight_layout()
plt.show()
```



```
In [22]: from metrics.statistic import compute_cov
print("Relative Frobenius Distance of Covariance Matrix")
print(f" {'Train':<16} {compute_cov(train_model.F.drop(columns=['alpha']).values, hist_stress):.4f}")
print(f" {'Diffusion':<16} {compute_cov(lower003, hist_stress):.4f}")
print(f" {'Gaussian':<16} {compute_cov(sampled_gaussian, hist_stress):.4f}")
```

```
corr_uncon    = corr_mat(uncon)
corr_rej      = corr_mat(rej)
corr_low      = corr_mat(lower003)
corr_gaussian = corr_mat(sampled_gaussian)
corr_train    = corr_mat(train_model.F.drop(columns=['alpha']).values)
```

Relative Frobenius Distance of Covariance Matrix

```
Train          0.5986
Diffusion      2.8553
Gaussian       0.7407
```

```
In [23]: from factor_model import load_model, reconstruct_returns
```

```
# --- load factor model ---
train_model = load_model(f"{PREFIX}")

# --- generated stress stock returns ---
fs_gen = np.column_stack([
```

```

        np.ones((len(lower003), 1), dtype=np.float32),
        lower003.astype(np.float32),
    ])
R_gen_stress = reconstruct_returns(train_model, fs_gen) # (N, S)

# --- historical stock returns on stress days ---
stress_mask = train_model.F["momentum"].values < q3[fidx]
fs_hist = train_model.F.values[stress_mask].astype(np.float32)
R_hist_stress = reconstruct_returns(train_model, fs_hist) # (T_stress, S)

train_vals = hist.values # (T, 7)

pca = PCA(n_components=2)
pca.fit(train_vals)

train_pc = pca.transform(train_vals)
uncon_pc = pca.transform(uncon)
con_pc = pca.transform(lower003)

xlab = f"PC 1 ({pca.explained_variance_ratio_[0]*100:.1f}%)"
ylab = f"PC 2 ({pca.explained_variance_ratio_[1]*100:.1f}%)"

N_SUB = 300

def _sub(arr, n=N_SUB, seed=0):
    idx = np.random.default_rng(seed).choice(len(arr), size=min(n, len(arr)))
    return arr[idx]

def _contour(ax, pc, color, levels=5, lw=1.2):
    kde = gaussian_kde(pc.T)
    pad = (pc.max(axis=0) - pc.min(axis=0)) * 0.15
    lo, hi = pc.min(axis=0) - pad, pc.max(axis=0) + pad
    xx, yy = np.mgrid[lo[0]:hi[0]:80j, lo[1]:hi[1]:80j]
    z = kde(np.vstack([xx.ravel(), yy.ravel()])).reshape(xx.shape)
    ax.contour(xx, yy, z, levels=levels, colors=[color], alpha=0.65, linewidth=1.2)

COLORS_PCA = {
    "train": "#888888",
    "uncon": "#4CAF50",
    "con": "#2196F3",
}

fig, ax = plt.subplots(figsize=(7, 5))
_contour(ax, train_pc, COLORS_PCA["train"])
_contour(ax, uncon_pc, COLORS_PCA["uncon"])
_contour(ax, con_pc, COLORS_PCA["con"])
ax.scatter(*_sub(train_pc, seed=0).T, c=COLORS_PCA["train"], marker=".", s=10)
ax.scatter(*_sub(uncon_pc, seed=1).T, c=COLORS_PCA["uncon"], marker="^", s=10)
ax.scatter(*_sub(con_pc, seed=2).T, c=COLORS_PCA["con"], marker="x", s=10)
ax.set_xlabel(xlab); ax.set_ylabel(ylab)
ax.set_title("PCA Projection of Factor Scenarios", fontweight="bold")
ax.legend(fontsize=9, markerscale=1.5)
ax.grid(True, alpha=0.25)

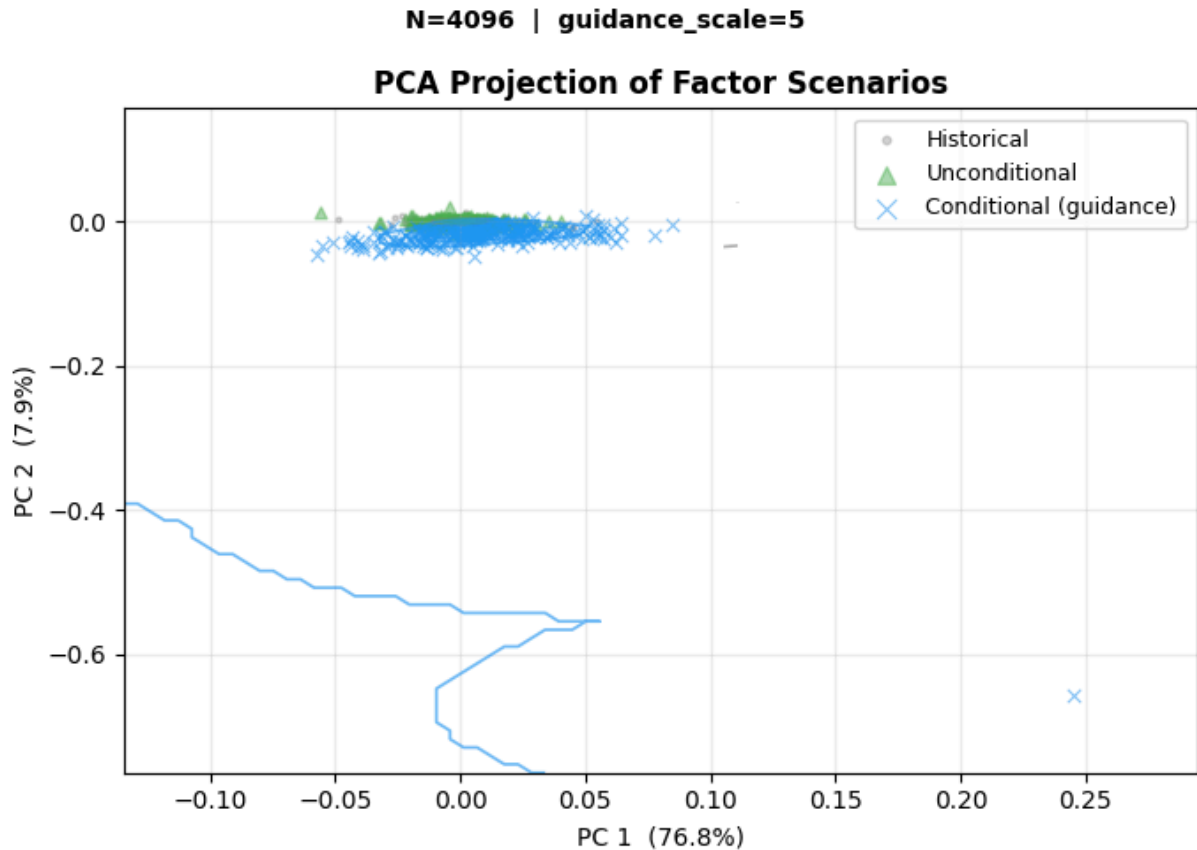
fig.suptitle(
    f"N={N} | guidance_scale={GUIDANCE_S}",

```

```

    fontsize=10, fontweight="bold"
)
plt.tight_layout()
plt.show()

```



```

In [24]: # Review.pdf 12.1.C.3 – Stock-level scatter: beta vs conditional expected re
from matplotlib.lines import Line2D

beta_cols      = list(train_model.F.columns)           # ["alpha", "market",
beta_mom_idx   = beta_cols.index("momentum")
beta_mom       = train_model.beta[beta_mom_idx, :]     # (S,)

# Reuse the conditional factor samples already drawn in earlier cells
# (lower003, sampled_gaussian) and reconstruct stock returns through train_m
def _to_returns(factors):
    fs = np.column_stack([np.ones((len(factors), 1), dtype=np.float32),
                           factors.astype(np.float32)])
    return reconstruct_returns(train_model, fs)

R_diff_cond    = _to_returns(lower003)
R_gauss_cond   = _to_returns(sampled_gaussian)

# Historical stress-day reconstruction (apples-to-apples)
stress_mask    = train_model.F["momentum"].values < q3[fidx]
R_hist_cond    = reconstruct_returns(
    train_model, train_model.F.values[stress_mask].astype(np.float32)
)

ER_diff       = R_diff_cond.mean(axis=0)

```

```

ER_gauss = R_gauss_cond.mean(axis=0)
ER_hist = R_hist_cond.mean(axis=0)

def _fit_with_ci(x, y, n_boot=300, seed=0):
    m, b = np.polyfit(x, y, 1)
    xs = np.linspace(x.min(), x.max(), 100)
    rng = np.random.default_rng(seed)
    n = len(x)
    preds = np.empty((n_boot, xs.size))
    for i in range(n_boot):
        idx = rng.integers(0, n, n)
        mi, bi = np.polyfit(x[idx], y[idx], 1)
        preds[i] = mi * xs + bi
    lo, hi = np.percentile(preds, [2.5, 97.5], axis=0)
    return xs, m * xs + b, m, lo, hi

series = [
    ("Diffusion", ER_diff, COLORS["con"], "o"),
    ("Gaussian", ER_gauss, COLORS["gaussian"], "^"),
    ("Historical stress", ER_hist, "#D62728", "x"),
]

fig, ax = plt.subplots(figsize=(8.5, 5.5), dpi=120)
ax.axhline(0, color="0.35", lw=0.6, alpha=0.5, zorder=1)

handles = []
for label, ER, color, mk in series:
    edge = color if mk == "x" else "white" # 'x' has no fill – edge carries
    ax.scatter(beta_mom, ER, s=34, alpha=0.55, color=color, marker=mk,
               edgecolors=edge, linewidths=0.7, zorder=3)
    xs, ys, slope, lo, hi = _fit_with_ci(beta_mom, ER)
    ax.fill_between(xs, lo, hi, color=color, alpha=0.13, zorder=2, linewidth=0.7)
    ax.plot(xs, ys, color=color, lw=2.0, alpha=0.95, zorder=4)
    handles.append(Line2D(
        [0], [0], color=color, lw=2, marker=mk, markersize=7,
        markerfacecolor="none" if mk == "x" else color,
        markeredgecolor=color if mk == "x" else "white",
        markeredgewidth=0.8,
        label=f"{label} (slope = {slope:+.4f})",
    ))

ax.set_xlabel(r"Stock momentum loading $\beta$", fontsize=11)
ax.set_ylabel("Conditional expected stock return", fontsize=11)
ax.set_title(
    f"Stock-level expected return under momentum stress "
    f"(momentum < {q3[fidx]:.4f})",
    fontsize=11,
)

ax.legend(handles=handles, fontsize=9, frameon=True, framealpha=0.92,
          loc="lower left", handlelength=2.6)
ax.grid(True, alpha=0.25, linestyle="--", linewidth=0.6)
ax.spines[["top", "right"]].set_visible(False)
plt.tight_layout()
plt.show()

```

/tmp/ipykernel_19232/2771927498.py:53: UserWarning: You passed a edgecolor/edgecolors ('#D62728') for an unfilled marker ('x'). Matplotlib is ignoring the edgecolor in favor of the facecolor. This behavior may change in the future.

```
ax.scatter(beta_mom, ER, s=34, alpha=0.55, color=color, marker=mk,
```

